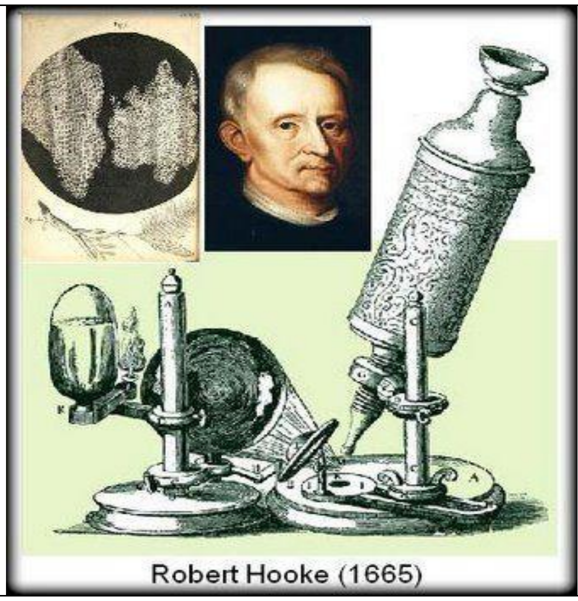
MICROSCOPY

Microscopy is the technique of using a microscope to observe objects that are too small to be seen by unaided/naked eye. It plays a crucial role in biology, medicine, materials science, and forensic investigations by allowing scientists to study microscopic structures, such as cells, bacteria, viruses, and nanoparticles.

History of microscopy

Microscopy developed gradually as scientists improved lenses and magnification instruments.

	
1. *Late 16th Century: The first compound microscope was developed around 1590 by Dutch spectacle makers Hans Janssen and Zacharias Janssen.*	2. *1665: Robert Hooke used a compound microscope to observe cork tissue and described tiny compartments which he called cells in his book Micrographia.*
	3. *1670s: Antonie van Leeuwenhoek built powerful simple microscopes and was the first to observe bacteria, protozoa, sperm cells, and red blood cells, laying the foundation of microbiology.*
4. *19th Century: Improvements in lenses and illumination increased clarity and enabled detailed study of cells, contributing to the development of the cell theory.*	

5. *19th Century: Improvements in lenses and illumination increased clarity and enabled detailed study of cells, contributing to the development of the cell theory.*	 
6. *20th Century: The electron microscope was developed, allowing scientists to observe extremely small structures such as viruses, organelles, and macromolecules.*	

Significance of microscopy in biology

Microscopy is extremely important in biology because it enables scientists to investigate the structure, organization, and activities of living organisms at the microscopic level.

1. Study of cells and cellular structure

Microscopy allows scientists to observe cells, which are the basic units of life. Through microscopes, biologists can examine cellular components such as the nucleus, cytoplasm, mitochondria, chloroplasts, and cell membranes, helping to understand how cells function and interact.

2. Discovery and study of microorganisms

Many living organisms such as bacteria, protozoa, algae, and fungi are too small to be seen with the naked eye. Microscopy made it possible to discover these organisms and study their structure, reproduction, and roles in ecosystems.

3. Medical diagnosis and disease investigation

Microscopes are widely used in medicine to examine blood samples, tissues, and body fluids. They help identify disease-causing organisms such as bacteria and parasites, detect abnormal cells such as cancer cells, and diagnose infections like malaria and tuberculosis.

4. Understanding cell division and development

Microscopy enables scientists to observe important biological processes such as mitosis and meiosis, which explain how cells grow, divide, and transmit genetic information from one generation to another.

5. Advancement of biological research

Modern biological research depends heavily on microscopy to investigate cellular and molecular processes. It has contributed greatly to fields such as genetics, microbiology, biotechnology, and molecular biology.

6. Environmental and ecological studies

Microscopes help scientists study microscopic organisms in soil, water, and air, allowing them to understand ecological relationships, nutrient cycles, and environmental changes.

7. Forensic and scientific investigations

Microscopy is used in forensic science to examine tiny pieces of evidence such as hair, fibers, pollen grains, and microorganisms, which can help solve crimes and investigate environmental contamination.

Key terms used in microscopy

1. Magnification

The ability of a microscope to increase the apparent size of an object. It is calculated as:

Total Magnification = Objective Lens magnification x Eyepiece Lens magnification

Example: If an objective lens is X40 and the eyepiece lens is X10, the total magnification is X400.

2. Resolution (Resolving power)

The ability to distinguish two close objects as separate entities. Higher resolution means a clearer and more detailed image.

Light microscopes have a resolution limit of approx. 0.2 μm, while electron microscopes can reach 0.1nm (TEM).

3. Contrast

The difference in brightness between the different parts of a specimen. Staining techniques, such as Gram staining and fluorescence staining, improve contrast.

4. Field of view (FOV)

The visible area seen through the microscope at a given magnification. As magnification increases, the field of view decreases.

5. Depth of field

The thickness of the specimen that remains in focus at one time. Higher magnification results in a shallower depth of field.

6. Numerical aperture (NA)

A measure of a lens's ability to gather light and resolve fine details. A higher NA value improves resolution.

7. Working distance

The distance between the objective lens and the specimen when it is in focus. Higher magnification lenses have a shorter working distance.

8. Parfocal

The ability of a microscope to remain in focus when switching between different objective lenses.

9. Staining

The use of dyes to highlight specific structures in a specimen.

Examples:

- ✓ Gram stain (for bacteria classification).
- ✓ Hematoxylin and eosin (H&E) stain (for tissue samples).

10. Immersion oil
A special oil used with high-magnification lenses (100× objective) to increase resolution by reducing light refraction.
11. Phase contrast
A technique that enhances contrast in unstained, live specimens, making internal structures more visible.
12. Fluorescence microscopy
Uses fluorescent dyes and UV light to highlight specific cell structures. Common in medical and genetic research.
13. Electron microscopy (EM)
Uses electron beams instead of light to achieve high magnification and resolution. Includes Transmission Electron Microscopy (TEM) and Scanning Electron Microscopy (SEM)

TYPES OF MICROSCOPES

There are two main types of microscopes: light microscopes and electron microscopes. These differ in terms of their working principles, resolution, magnification, and applications.

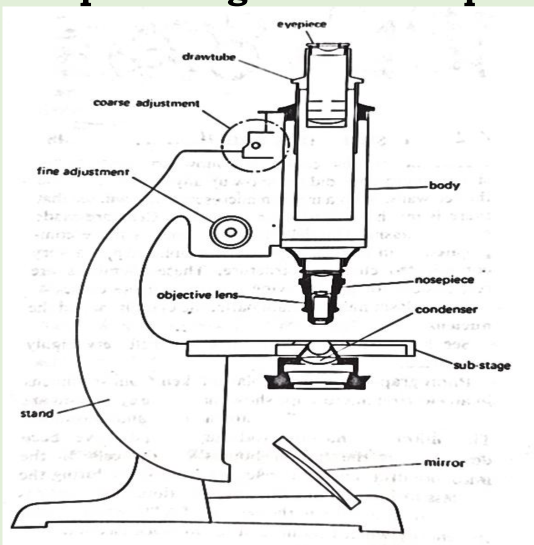
1. LIGHT MICROSCOPE
A light microscope (also called an optical microscope) uses visible light to illuminate and magnify specimens. It is commonly used in schools, universities, and research laboratories for examining biological tissues, microorganisms, and cells.

Functions of the Parts of a Light Microscope

1. Eyepiece (Ocular lens)
Magnifies the image formed by the objective lens so that it can be viewed by the observer. It usually provides ×10 magnification.

2. Draw tube (Body tube)
Holds the eyepiece and maintains the correct distance and alignment between the eyepiece and objective lenses.

Structure of a modern compound light microscope



3. Coarse adjustment knob: Moves the body tube or stage up and down rapidly to bring the specimen into rough focus, especially when using low power.
4. Fine adjustment knob: Moves the body tube or stage slowly and precisely to obtain a clear and sharp image, particularly when using high power.
5. Body: Supports the upper parts of the microscope and connects the eyepiece to the objective lenses, ensuring proper alignment of the optical system.
6. Nosepiece (Revolving nosepiece): Holds the objective lenses and allows them to be rotated so that different magnifications can be used.
7. Objective lens: The main magnifying lenses of the microscope. They magnify the specimen at different powers such as ×4, ×10, ×40, or ×100.
8. Condenser: Concentrates and focuses light onto the specimen to improve illumination and clarity of the image.
9. Substage: Supports the condenser and diaphragm and helps control the amount of light passing to the specimen.
10. Mirror: Reflects light from an external source upwards through the condenser and specimen to illuminate it.
11. Stand (Arm and base): Supports the entire microscope and provides stability. The arm is also used when carrying the microscope.

Principle of operation of the compound light microscope

A light microscope operates based on the principle of using visible light to illuminate a specimen and magnify its image through a series of glass lenses. The key steps involved in its operation include the following:

1. Specimen preparation: The specimen must be thin for light to penetrate through it, this can be don by slicing though the specimen to obtain a thin layer or by peeling a thin layer from a specimen.

Staining. with dyes (e.g., methylene blue, iodine) enhances contrast and highlights structures of interest.

2. Illumination: A light source illuminates the specimen, while the condenser lens focuses the light onto the specimen for better visibility.
3. Magnification by the Lenses: The objective lens enlarges the specimen's real image, which the eyepiece lens further magnifies into a larger virtual image.

Total magnification is calculated as:
Total Magnification = Objective lens magnification × Eyepiece lens magnification

4. Image Formation: The light is transmitted through the specimen and refracted (bent) by the lenses in order to focus the image. The final magnified image appears inverted and enlarged.
5. Resolution and Contrast: The resolution of a light microscope is limited to around 200 nm due to the wavelength of visible light. Contrast can be improved by staining the specimen or adjusting the diaphragm and condenser settings.

Types of light microscopes

- **a)** Bright-Field Microscope: The most common type, where light passes through the specimen. May require staining to enhance visibility.
- **b)** Phase-Contrast Microscope: Enhances contrast without staining, useful for live cells.
- **c)** Fluorescence Microscope: Uses fluorescent dyes and UV light to highlight specific structures.
- **d)** Confocal Laser Scanning Microscope: Uses laser beams to produce high-resolution 3D images of specimens.

Advantages and limitations of light microscopes

Advantages of using a light microscope

- ❖ Simple and easy to use since it requires minimal training and is widely available in schools, laboratories, and research centres.
- ❖ Can observe living cells and hence allow the study of live specimens in real-time unlike electron microscopes.
- ❖ Relatively inexpensive compared to the electron microscopes, they are affordable and require less maintenance.
- ❖ Uses natural or artificial light and hence no need for a vacuum or complex preparation techniques.

Limitations of a light microscope

- ❖ Limited magnification, maximum magnification is between 1000× to 2000×, making it unsuitable for observing very small structures like organelles in detail.

- ❖ Low resolution, the resolving power is about 0.2 µm (200 nm), which limits the ability to distinguish fine details in cells and microorganisms.
- ❖ Limited contrast, requires staining to enhance visibility, which may sometimes alter specimen properties.

2. ELECTRON MICROSCOPE

An electron microscope (EM) uses a beam of electrons instead of light to achieve extremely high magnification and resolution. It allows for detailed imaging of cellular ultrastructure, including organelles and molecules.

Principle of Operation of an electron microscope

- A beam of electrons is directed at the specimen.
- Instead of glass lenses, electromagnetic lenses focus the electron beam.
- The interaction between electrons and the specimen creates highly detailed images.

Types of electron microscopes

a) Transmission Electron Microscope (TEM)
  - ✓ Electrons pass through a thin specimen and can produce 2D, high-resolution images of internal cell structures like organelles.
  - ✓ Magnification of up to 2,000,000× and Resolution of about 0.1 nm (nanometres)

b) Scanning Electron Microscope (SEM)
  - ✓ Electrons bounce off the surface of the specimen producing 3D images of surface structures.
  - ✓ Magnification of up to 500,000× and resolution of about 0.5 nm
  - ✓ Used for examining cell surfaces, viruses, and bacteria.

<u>Advantages and Disadvantages of Electron Microscopes</u>

<u>Advantages of a light microscope</u>

- ❖ Extremely high magnification and resolution. Electron microscopes can magnify up to 2,000,000x, with a resolution as low as 0.1 nm, allowing for detailed imaging of cellular structures.
- ❖ Reveals ultrastructure of cells, unlike light microscopes, electron microscopes can visualize ribosomes, viruses, organelles, and molecular structures in great detail.

- Different imaging techniques; Transmission Electron Microscopes (TEM) provide highly detailed internal structures, while Scanning Electron Microscopes (SEM) offer 3D surface images.

Limitations of an electron microscope

- Electron microscopes are costly to purchase and maintain, requiring specialized facilities and high operational costs.
- Specimens must be dead due to the vacuum conditions required for electron beam transmission, living specimens cannot be observed.
- Large and requires specialized training; unlike the light microscopes, electron microscopes are bulky and complex and need skilled personnel for operation and specimen preparation.
- Time-consuming sample preparation since specimens require complex processing, including dehydration, fixation, and staining with heavy metals to enhance contrast.

Comparison between a light microscope and an electron microscope

Feature	**Light microscope**	**Electron Microscope**
Radiation source	Visible light	Electron beam
Lenses	Glass lenses	Electromagnetic lenses
Resolution	Approx. 0.2 µm	Approx. 0.1 nm (TEM), Approx. 0.5 nm (SEM)
Magnification	1000x - 2000x	Up to 2,000,000x (TEM), 500,000x (SEM)
Specimen preparation	Simple, can use live specimens	Requires special preparation, specimen must be dead
Image type	Colour images	Black and white
Observation	Whole cells, tissues	Cell ultrastructure, viruses, organelles
Cost and complexity	Cheap and easy to use	Expensive, requires specialized training

Applications of microscopy

Here we compare the uses of light microscopy and electron microscopy across different fields.

Field	Light Microscopy Uses	Electron Microscopy Uses
Biology	Observing cells, tissues, and microorganisms such as bacteria, fungi, and protozoa.	Studying ultrastructures like viruses, organelles (mitochondria, ribosomes), and molecular details of cells.
Medicine	Diagnosing diseases such as malaria, tuberculosis (TB), and fungal infections through stained slides.	Identifying pathogens (e.g., viruses, bacteria) at the nanometre scale, crucial for virology and medical research.
Forensic science	Examining hair, blood cells, textile fibres, and tissue samples to identify suspects and causes of death.	Analysing gunshot residues, micro-particles, and forensic trace evidence with high resolution.
Nanotechnology	Basic examination of materials and surface structures at the micron level.	Studying nanoparticles, molecular arrangements, and nanoscale materials used in advanced technology.
Archaeology	Identifying ancient pollen, plant cells, and biological remains on artifacts.	Examining ancient tools, artifacts, and microstructures to determine age, composition, and authenticity.

Note

- Light microscopes are commonly used for live cell studies, medical diagnosis, and forensic analysis.
- Electron microscopes provide greater detail and are essential for virology, nanotechnology, and material sciences, but require dead specimens and complex preparation.

The concept of magnification

Magnification is the increase in the apparent size of an object without changing its actual size.
Or the process of making an object appear larger than its actual size using optical or digital methods.
In microscopy, magnification refers to the process of enlarging the appearance of a specimen using lenses of the microscope.

The amount of enlargement is expressed as a numerical value called "magnification." When this number is less than one, it indicates a size reduction, also known as minification or de-magnification.
A magnification number greater than 1 means that the image appears larger than the actual object. For example, a magnification of 10x or (X10) means the object appears ten times larger than its real size. This enlargement helps in observing fine details that would otherwise be invisible to the naked eye.

Magnification power

The magnification power of a lens refers to how much larger an image may look when seen through that lens.

Linear magnification

Linear magnification (sometimes called lateral or transverse) magnification refers to the ratio of image length to object length measured in planes that are perpendicular to the optical axis.
Or simply how much bigger (or smaller) an image appears compared to the actual object, based on the length in a specific direction (usually side to side).

Formula: Linear magnification **(M) =** $\mathbf{\dfrac{Image\ size\ (Length)}{Object\ size\ (Length)}}$

If the magnification is greater than 1, the image is larger than the object. If it's less than 1, the image is smaller than the object.

Example:

Imagine you are observing a leg of housefly through a microscope then after you make its drawing/ image on paper and you want to determine the linear magnification.

- Object Length: The actual size of the leg is 1 cm.
- Image Length: The drawing (Image) of the leg on paper is 4 cm long.

Now, calculate the **linear magnification**:

Linear Magnification **(M)=** $\mathbf{\dfrac{4\ cm}{1\ cm} = X4}$

This means the image/ drawing of the leg of a housefly is 4 times larger than its actual size.
If the magnification were negative, like **-4**, it would mean the image is inverted (flipped upside down) compared to the real insect.
A negative value for magnification means the image is inverted (flipped upside down).

The field of view (FOV) of a microscope

The field of view (FOV) of a microscope refers to the visible area you can observe through the lens system at a given time. It is typically measured in terms of the ***diameter of the circle*** you can see when you look through the microscope.
The size of the field of view depends on factors like:

1. Objective lens magnification: Higher magnifications (e.g., 100x) result in a smaller field of view but reveals more detail compared to lower magnifications with large field of view, less detail (e.g., 4x).
2. Eyepiece: Different eyepieces may have different field of view sizes.
3. Optical design: The design of the microscope's optics can influence how much of the specimen is visible at once.

Measuring the field of view:

a) Using a stage micrometre

- ❖ Place a stage micrometre (a slide with a known scale) on the microscope stage.
- ❖ Calibrate the eyepiece graticule (a scale in the eyepiece) against the stage micrometre.
- ❖ Determine the value of each division on the eyepiece graticule in micrometres (μm) by comparing it to the known scale on the stage micrometre.
- ❖ The field of view (FOV) diameter is the distance across the circular area you see through the microscope.
- ❖ For low magnification objectives, you may need to move the 1 mm scale several times to determine the overall diameter of the field of view in mm, which is then multiplied by 1000 to convert to microns.

b) Using a ruler to measure the field of view

- ✓ Place a clear metric ruler on the microscope stage beside the slide to measure the field of view. Ensure that the ruler is

aligned properly, and that the scale is visible for accurate measurement.
- ✓ Set the microscope to the lowest magnification (usually 4x or 10x). Adjust the focus to get a clear image of the field of view through the eyepiece.
- ✓ Look through the eyepiece and observe the diameter of the visible area (the field of view).
- ✓ Align the ruler along the edge of the field of view and measure the diameter of the field of view in millimetres (mm) by noting the length across the visible area.
- ✓ Record the field diameter in millimetres (mm) for reference.
- ✓ To convert this measurement into micrometres (μm), multiply the field diameter in millimetres by 1000 (since 1 mm = 1000 μm).

$$\textit{Field of View } (\mu m) = \textit{Field of View } (mm) \times 1000$$

Example:
If the field of view is 0.2mm, the conversion would be:
0.2 mm x1000=200 μm

Once you have the field diameter in micrometres (μm), use it to estimate the linear magnification of the cells you observe using the following formula:

$$\textit{Magnification} = \frac{\textit{Field of view at low power}}{\textit{Size of the cell at low power}}$$

Calculating field of view (FOV) diameter
As magnification increases, the diameter of the field of view decreases. This is because higher magnification allows you to see a smaller area in greater detail.
The formula to calculate the field of view diameter is:

$$\textit{Diameter of the field of view (mm)} = \frac{F}{M}$$

where:
- **F** = the field number (a value based on the eyepiece).
- **M** = the magnification of the objective lens.

If you know the field diameter at the lowest magnification, you can calculate the field diameter at higher magnifications.

Example

Measure the Field of View at Low Magnification: For example, the field diameter at low magnification (4x) is 2 mm.

Use the Formula: Suppose the field number (F) of your eyepiece is **18** (this is a standard value that can vary depending on the microscope) and the magnification (M) of the objective lens is **10x.**

To calculate the field diameter at 10x magnification, use the formula:

$$\textit{Diameter of the field of view (mm)} = 18 \div 10 = 1.8 \text{ mm}$$

So, the field diameter at 10x magnification is 1.8 mm.

Generally; As the magnification increases, the field of view diameter decreases. This means that under higher magnifications, you will be able to see less of the specimen at one time, but in more detail.
For example, if the diameter at 4x magnification was 2 mm, then at 40x magnification, it will decrease to 0.2 mm.
Therefore; Using the field number and magnification, you can calculate the size of the field of view at different magnifications.

Estimating cell size

a) Using the calibrated eyepiece graticule

- ❖ Once the eyepiece graticule is calibrated, you can use it to measure the length or width of cells.
- ❖ Count the number of graticule divisions that correspond to the cell's size and multiply by the value of one graticule division in micrometres.

b) Estimating by dividing the FOV diameter

- ❖ Measure the FOV diameter: The diameter of the field of view is given in micrometres (μm), for example, 5000 μm (5 mm).
- ❖ Estimate by counting how many cells, laid end-to-end, fit across the diameter of the field of view.
- ❖ Divide the field of view diameter (in micrometres) by the number of cells that fit across to get an estimate of the average cell size.
- ❖ For example, if the field diameter is 5 mm (5000 μm) and you count a total of 50 cells laid end-to-end across the diameter of the field, then

$$\textbf{Cell size} = \frac{\textit{FOV diameter}}{\textit{Number of cells}} = \frac{5000\ \mu m}{50\ \text{cells}} = 100\ \mu m \text{ per cell}$$

The cell size is just an estimate because the cells are not of the same size and shapes

Calculating the magnification of a drawing of an onion cell as observed from microscope.

Given measurements:

- Size of the drawing as measured by the ruler **= 50 mm**
- Actual size of the onion cell as obtained in microscope =100 **µm**

Convert the size of the drawing to micrometres (50 ÷ 0.001) = 50,000 **µm**

Use the magnification formula:

$$\textit{Magnification} = \frac{\text{Size of drawing (mm)}}{\text{Actual size of the cell (mm)}}$$

$$\textit{Magnification} = \frac{50{,}000\ \mu m}{100\ \mu m}$$

$$\textit{Magnification} = 500$$

The magnification of the onion cell drawing is X500 (500 times).

Remember: If the ruler used is measured in millimetres (mm), we will convert all measurements to micrometres (µm) for consistency.

Metric conversions necessary for your microscopic computations

Table Showing Metric Conversions for Length

Unit	Symbol	Factor	Equivalent
Meter	m	1	1 m = 1 meter
Decimetre	dm	10^{-1} m	1 dm = 0.1 meter
Centimetre	cm	10^{-2} m	1 cm = 0.01 meter
Millimetre	mm	10^{-3} m	1 mm = 0.001 meter
Micrometre	µm	10^{-6} m	1 µm = 0.000001 meter (1 millionth of a meter)

Length (Using a centimetre as the standard)

Unit	Symbol	Factor	Equivalent
Centimetre	cm	1	1 cm = 1 cm
Millimetre	mm	10^{-1} cm	1 mm = 0.1 cm
Micrometre	µm	10^{-2} cm	1 µm = 0.01 cm

millimetre (mm) as the standard unit for length conversions within the metric system.

Unit	Symbol	Factor	Equivalent
Centimetre	cm	10^1 mm	1 cm = 10 mm
Millimetre	mm	1	1 mm = 1 mm
Micrometre	µm	10^{-3} mm	1 µm = 0.001 mm

The **micrometre (µm)** is commonly used to measure small distances, such as the diameter of cells or microscopic organisms.

STAINING TECHNIQUES AND SAMPLE PREPARATION IN MICROSCOPY

Staining is a technique used in microscopy to enhance contrast in biological specimens, making specific structures more visible under the microscope. Different stains interact with various cellular components, allowing researchers to differentiate organelles, bacteria, and tissue types.
Sample preparation is crucial before staining to ensure the specimen is preserved and visualized correctly. The preparation process varies depending on the type of microscope being used (light vs. electron microscopy) and the type of specimen (e.g., bacteria, tissues, cells).

General steps in sample preparation for staining

The process of preparing a biological sample for staining includes:
(a) Fixation: Preserves the specimen's structure and prevents decomposition. Formaldehyde is the commonly used fixatives for light microscopy.
(b) Dehydration: Removes water from the sample by use of increasing concentrations of ethanol or acetone.
(c) Embedding: The sample is embedded in a medium like paraffin wax (light microscopy) or resin (electron microscopy) to provide support for slicing.
(d) Sectioning: The sample is cut into thin slices using a microtome (for light microscopy) the typical thickness for light microscopy is 4-10 µm
(e) Staining: The specimen is treated with a stain to highlight specific components where different stains target nuclei, cytoplasm, organelles, and bacteria.
(f) Mounting: The stained sample is placed on a slide, covered with a coverslip, and sealed with mounting medium.

NB: In cases where steps **(c)** and **(d)** are not possible, the learner should just ensure that the specimen is slices ore peeled as thin as possible to allow more light to go through it as this will increase visibility of the underlying structures.

Types of staining techniques

1. Simple staining: Uses a single stain to colour cells/ structures uniformly.

Example: Methylene blue stain (for bacteria).

2. Differential staining: Uses two or more dyes to distinguish different cell types or structures.

Examples: Gram Staining (for bacterial classification) and Acid-Fast Staining (for Mycobacterium tuberculosis).

3. Special staining: Targets specific structures like capsules, flagella, or spores.

Examples: Capsule Stain (for bacterial capsules) and the Endospore Stain (for spore-forming bacteria).

4. Fluorescent staining: Uses fluorescent dyes that emit light under UV.

Example: DAPI (*4,6-Diamidino-2-phenylindole*) stain (binds to DNA, used in cell biology).

Common stains used in microscopy

Stain	**What it Shows/stains**	**Appearance**	**Use/Applications**
Methylene blue	Nuclei and cytoplasm of cells	Blue	Used in biology to highlight cell structures and in microbiology.
Gram's Stain	Differentiates bacteria (Gram-positive vs. Gram-negative)	Purple for Gram-positive, Pink for Gram-negative	Identifies bacterial species and classifies them.
Iodine stain	Starch granules in plant cells	Blue-black	Used in plant biology to detect starch in plant tissues.
Safranin	Cell walls, nuclei, and cytoplasm	Red	Counterstain in Gram staining, used for plant tissue studies.
Crystal violet	Bacterial cells	Purple	Primary stain in Gram staining to differentiate bacterial species.

Eosin	Cytoplasm, extracellular matrix, and some cell structures	Pink	Used in histology to highlight animal tissue structures.
Hematoxylin	Nuclei, chromatin, and nucleic acids	Blue or purple	Common in histology to stain tissue sections, especially for DNA.
Alcian blue	Acidic polysaccharides, such as glycosaminoglycans	Blue	Used to detect mucus, cartilage, and acidic substances in tissues.
Trypan blue	Dead cells	Blue	Vital stain to distinguish living from dead cells.
Neutral red	Lysosomes, vacuoles, and cell organelles	Red	Stains acidic organelles, often used in live cell studies.

Importance of staining and sample preparation

- Enhances contrast in specimens for better visualization.
- Distinguishes different cell types (bacteria, fungi, human cells).
- Identifies pathogens in medical diagnostics.
- Reveals cellular structures (nucleus, mitochondria, membranes).
- Essential for histology, pathology, and microbiology research.

<u>MICROSCOPIC OBSERVATION OF PLANT, ANIMAL, AND BACTERIAL CELLS.</u>

1. Skin tissue (Epidermis and dermis)

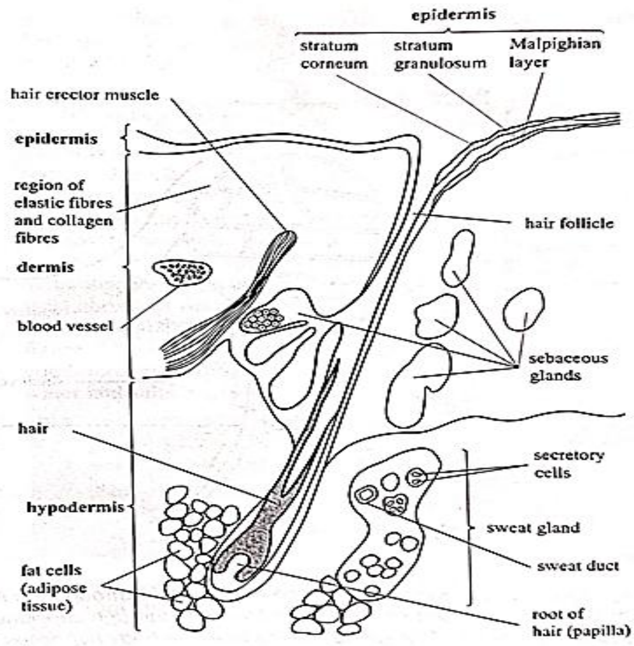
Aim: To observe the epidermis, dermis, and structures like hair follicles.

Procedure

1. Obtain a small section of rat skin using sharp blade, take care to avoid cuts and injury.
2. Fix the tissue in 10% Neutral Buffered Formalin at a ratio of 10:1 for 20 minutes to preserve its structure for histological analysis. Or use tissue that has been fixed for 6-24 hours.
3. Cut/peel thin sections of the tissue, float them in a warm water bath, mount on glass slides, and dry for staining and analysis.
4. Use Hematoxylin and Eosin (H&E) to visualize the epidermis (outer layer), dermis (inner layer), and hair *follicles.*
5. Mount the tissue on a slide and cover with a coverslip.

Observation

Observe the epidermal layers, dermal structures, and hair follicles.



<u>Low Power (4x or 10x)</u>

Epidermis appears as a thin, continuous outer layer, slightly darker than the underlying dermis; *the dermis is thicker and composed of loosely packed connective tissue; hair follicles are visible as small oval or elongated structures embedded in the dermis;* *the overall structure of the skin is distinguishable, but finer details such as individual cell layers are not clear.*

<u>Medium power (40x)</u>

The epidermis reveals multiple layers, with a distinct stratified squamous structure; the outermost stratum corneum appears as a thin, keratinized layer; the dermis contains connective tissue with visible fibroblasts, blood vessels, and sebaceous glands; hair follicles more defined, showing their structural orientation within the dermis; sweat glands may be visible as coiled structures in the dermal layer.

<u>High Power (100x, Oil Immersion)</u>

Individual epidermal cells are visible, including keratinocytes and basal layer cells; the dermo-epidermal junction is clearly defined, showing attachment points between the two layers; hair follicles show cellular details, including the hair matrix and associated sebaceous glands; blood vessels, collagen fibres, and fibroblast nuclei are clearly visible within the dermis; some immune cells such as mast cells may be identified within the connective tissue.

2. Muscle tissue (Skeletal muscle)

Aim: To observe muscle fibres, striations, and nuclei.

Procedure

1) Obtain a small section of rat skin using sharp blade, take care to avoid cuts and injury.
2) Fix the tissue in 10**%** Neutral Buffered Formalin at a ratio of 10:1 for 20 minutes to preserve its structure for histological analysis. Or use tissue that has been fixed for 6-24 hours.
3) Cut/peel thin sections of the tissue, float them in a warm water bath, mount on glass slides, and dry for staining and analysis.
4) Stain with Hematoxylin and Eosin (H&E) to visualize striated muscle fibres and the multinucleated fibres.
5) Mount the tissue on a slide and cover with a coverslip.

Observation:

<u>Low Power (4x or 10x)</u>

Skeletal muscle appears long, parallel fibres arranged in bundles; the tissue is densely packed, with clear separation between muscle bundles by connective tissue (perimysium); no clear details of striations or nuclei are visible at this magnification; blood vessels and connective tissue structures observed surrounding the muscle fibres.

<u>Medium Power (40x)</u>

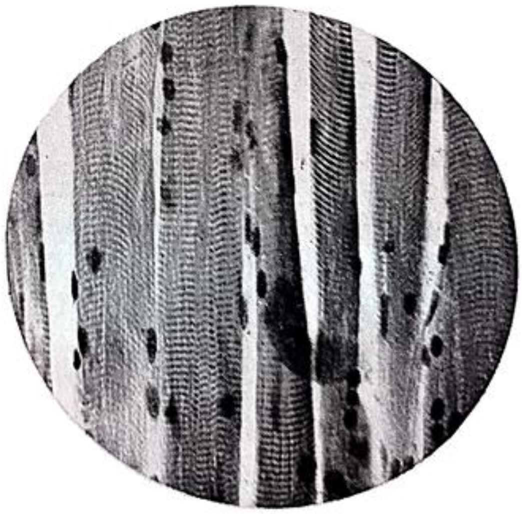
Individual muscle fibres (muscle cells or myocytes) become more distinct, appearing elongated and cylindrical; striations (alternating light and dark bands) start to be visible along the fibres, indicating the arrangement of actin and myosin filaments; nuclei are elongated and located at the periphery of the muscle fibres <u>(a distinguishing feature of skeletal muscle)</u>; some connective tissue layers (endomysium around individual fibres and perimysium around bundles) can be seen more clearly.

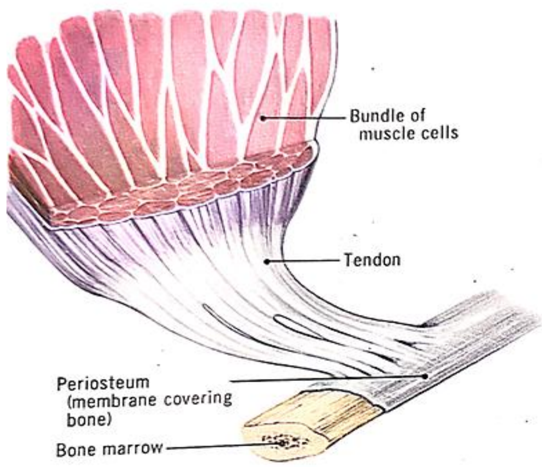
<u>High Power (100x, Oil Immersion)</u>

Striations are prominent, showing alternating dark (A bands) and light (I bands) regions within muscle fibres; the sarcomeres, which are the functional units of muscle contraction, are identifiable in well-prepared sections; multiple nuclei are clearly seen at the periphery of the fibres, confirming the multinucleated nature of skeletal muscle; fine details of connective tissue, such as endomysium surrounding individual muscle fibres, are more evident.

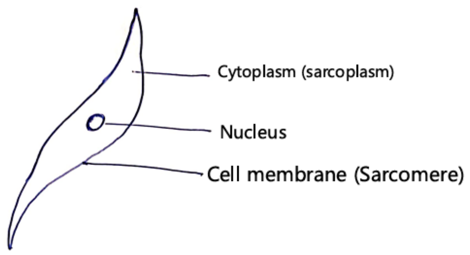
Function of the skeletal muscle

- ✓ The main function of skeletal muscle is to facilitate *voluntary movement* by contracting and generating force, allowing the body to perform actions such as walking, running, lifting, and maintaining posture.
- ✓ Skeletal muscles also contribute to *heat production*, *joint stability*, and *support* of the skeletal system by attaching to bones via tendons and enabling coordinated movement.

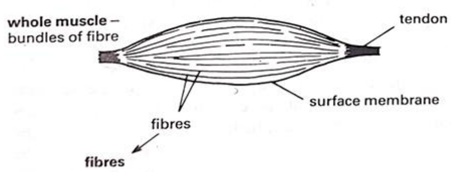

Skeletal muscle tissue

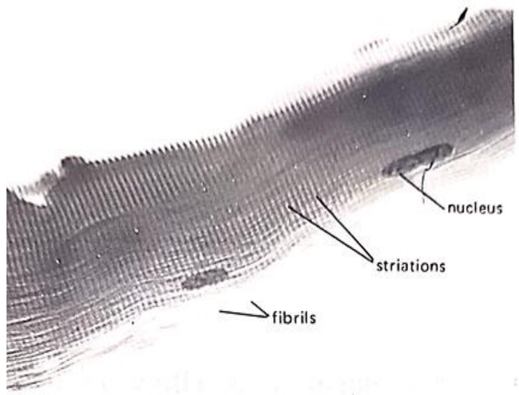


Muscle tissue held to the bone by a tendon



Skeletal muscle cell Structure





Adaptations of the skeletal muscle to function

- ❖ The distinct striations observed under the microscope, formed by alternating A-bands (dark) and I-bands (light), indicate the precise arrangement of actin and myosin filaments, which enables efficient muscle contraction.

- ❖ The presence of multiple nuclei located at the periphery of each skeletal muscle fibber allows for rapid protein synthesis and efficient coordination of metabolic activities required for muscle function and repair.
- ❖ The elongated, cylindrical shape of skeletal muscle fibres allows for the uniform transmission of contraction forces along the entire length of the muscle, enabling powerful and coordinated movement.
- ❖ The connective tissue layers surrounding muscle fibres, including the endomysium, perimysium, and epimysium, provide structural support, prevent overstretching, and facilitate the even distribution of contraction forces throughout the muscle.
- ❖ The dense network of blood vessels within skeletal muscle tissue ensures a continuous supply of oxygen and nutrients, which are essential for ATP production and sustained muscle contractions.
- ❖ The high concentration of mitochondria within skeletal muscle fibres supports the rapid production of ATP, which is necessary to power repetitive and sustained contractions during voluntary movements.

4. Blood tissue (Blood smear)

Aim: To observe red blood cells (RBCs), white blood cells (WBCs), and platelets.

Procedure

1. Place a drop of rat blood on a clean glass slide and spread it evenly using another slide to create a thin smear.
2. Allow the smear to air dry, then fix it by applying methanol for a few minutes.
3. Apply Wright's stain or Giemsa stain to differentiate red blood cells (RBCs), white blood cells (WBCs), and platelets.
4. Let the stain sit for the recommended time, then rinse gently with distilled water for Wright's stain, Let it sit for 1-3 minutes and Giemsa stain for 10-15 minutes (diluted stain) or 3-5 minutes (concentrated stain)
5. Dry the slide and cover the stained smear with a coverslip for observation under a microscope.

Observation under different magnifications

Low Power (10× Objective up to 100× Total Magnification)

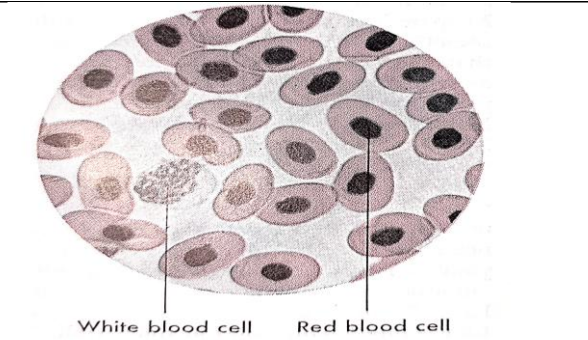
- ❖ RBCs: Seen as numerous small, pink-stained cells, uniformly distributed.
- ❖ WBCs: Appear as larger, scattered cells with darker-stained nuclei.
- ❖ Platelets: Hard to distinguish clearly but may appear as tiny specks.

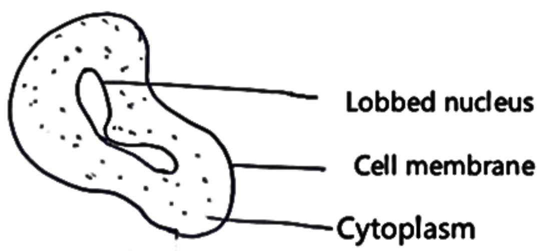
Medium Power (40× Objective up to 400× Total Magnification)

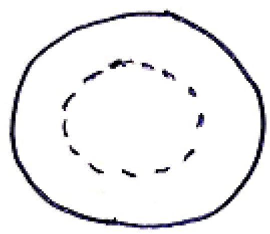
- ❖ RBCs: More distinct biconcave shape but still no internal details visible.
- ❖ WBCs: More visible; their nuclei and cytoplasmic granules can to be seen.
- ❖ Platelets: Seen as small, irregularly shaped fragments.

High Power (100× Oil Immersion, up to 1000× Total Magnification)

- ❖ RBCs: Clearly defined biconcave discs, uniform in shape and colour.
- ❖ WBCs: Detailed view of nucleus and cytoplasm, allowing differentiation of neutrophils, lymphocytes, monocytes, eosinophils, and basophils.
- ❖ Platelets: Clearly visible as tiny, dark-stained cell fragments involved in clotting.

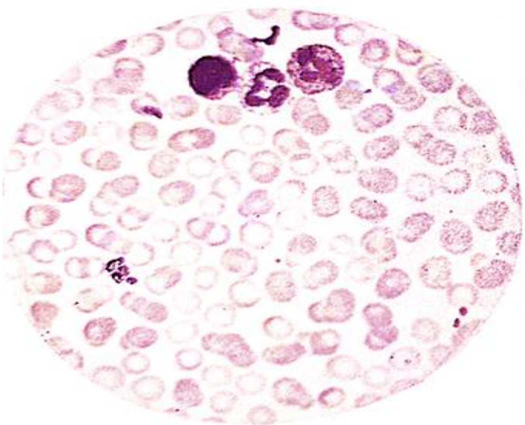
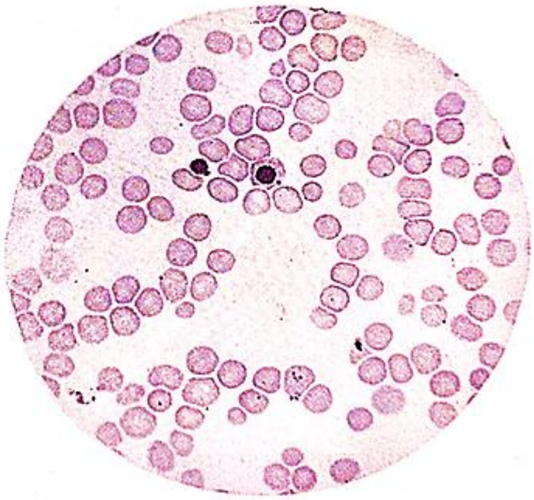






Adaptations of blood based on microscopic structures

Blood Component	**Structural Adaptation**	**Function**
Red Blood Cells (Erythrocytes)	Biconcave shape	Increases surface area for efficient gas exchange (oxygen and carbon dioxide).
	Absence of a nucleus	Provides more space for haemoglobin, maximizing oxygen transport.
	Flexible membrane	Allows red blood cells to squeeze through narrow capillaries for efficient oxygen delivery.
White Blood Cells (Leukocytes)	Presence of a nucleus	Enables production of enzymes and antibodies for immune defence.
	Ability to change shape (Amoeboid movement)	Allows movement through blood vessel walls to engulf pathogens and remove dead cells.
Platelets (Thrombocytes) Plasma	Irregular shape with sticky surfaces	Helps in blood clotting by adhering to damaged blood vessels and forming clots.
	Fluid medium	Facilitates the transport of nutrients, hormones, waste products, and blood cells throughout the body.

A person with normal blood count (X600)	A person with anaemia
	

A person homozygous for sickle cell anaemia X600 *(J. Hopkins u*

4. Spinal cord tissue

Aim: To observe nerve cells (neurons) and glial cells using compound light microscope.

Procedure

1) Obtain a small piece of rat spinal cord tissue.
2) Immerse the tissue in formalin to preserve its structure. This prevents degradation and helps maintain the cells' shape
3) After fixation, prepare thin slices (sections) of the tissue, thin sections allow the light microscope to focus clearly on the individual cells.
4) Apply the stain to the tissue sections, allowing it to sit for 1-3 minutes (for Hematoxylin) and 2-5 minutes (for Eosin). Rinse gently with distilled water to remove excess stain.
5) Hematoxylin stains the nuclei of cells a dark blue or purple, Eosin stains the cytoplasm and extracellular matrix in shades of pink or red.
6) After staining, carefully place the stained tissue section onto a glass slide. Apply a coverslip over the tissue to protect it and prevent contamination.

Observation under different magnifications

<u>Low Power (10× Objective, 100× Total Magnification):</u>

- *At low magnification, you will observe the overall structure of the spinal cord tissue, with scattered groups of cells.*
- *Neurons will be visible as larger cells, and glial cells will appear as smaller, scattered cells around the neurons. The general tissue architecture will be clear, but fine details are not distinguishable.*

<u>Medium Power (40× Objective, 400× Total Magnification):</u>

- *At medium magnification, you can begin to clearly distinguish individual neurons with their prominent nuclei and processes (axon and dendrites).*
- *Glial cells will be more defined, with visible, smaller nuclei. The arrangement and relative positions of neurons and glial cells will be clearer.*

<u>High Power (100× Oil Immersion, 1000× Total Magnification):</u>

- *At high magnification, you can observe fine details of neurons such as the nucleus, cytoplasm, and long processes (axons and dendrites).*
- *Glial cells will appear distinctly with smaller nuclei and their finer structural features, such as their supporting role around neurons. The fine structures within the cells, including any cytoplasmic features, will be visible.*

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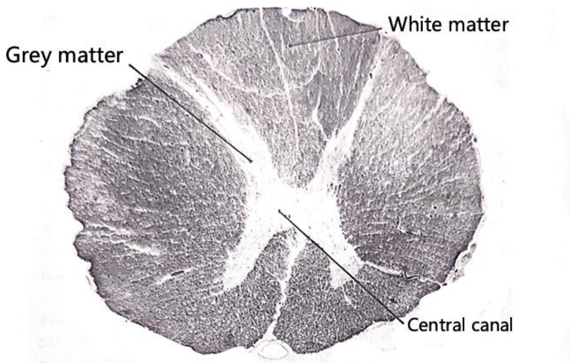
<tr><td>

Transverse section through the spinal cord</td><td></td></tr>
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5. Intestinal Tissue (Small intestine)

Aim: To examine the microscopic structure of the small intestine, focusing on villi, epithelial cells, and goblet cells at different magnifications.

Procedure

1) Obtain a section of the rat small intestine.
2) Preserve the tissue in formalin to prevent decomposition and maintain cellular structure.
3) Cut thin tissue sections for microscopic examination.
4) Apply Hematoxylin and Eosin (H&E) stain to highlight the villi and epithelial cells.

5) Use Periodic Acid-Schiff (PAS) stain to specifically identify goblet cells, which produce mucus.
6) Place the stained tissue section on a glass slide and cover it with a coverslip for observation.

Observation at different magnifications

<u>Low Power (4x or 10x Objective Lens)</u>

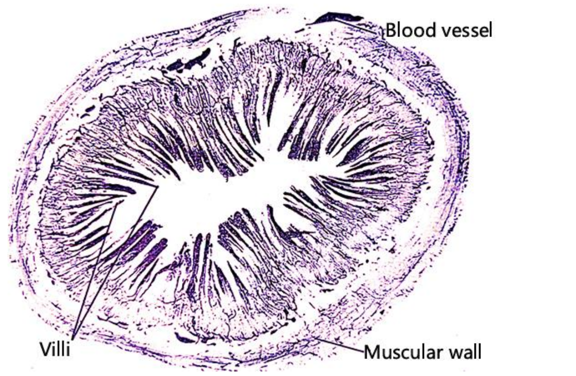
Identify the overall structure of the small intestine, including the arrangement of villi; general organization of the intestinal wall layers;

<u>Medium Power (20x or 40x Objective Lens)</u>

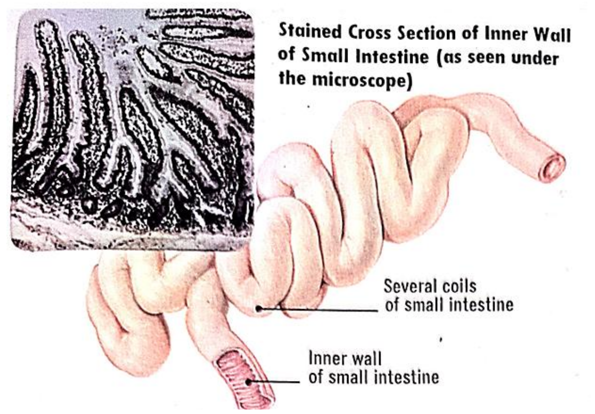
View individual villi in more detail; epithelial cells lining the villi' goblet cells; mucus-filled spaces among epithelial cells;

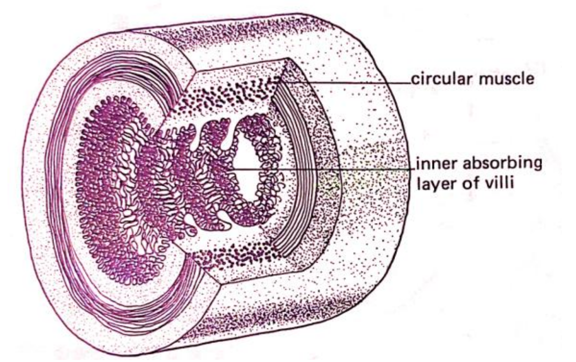
<u>High Power (100x Oil Immersion Objective Lens)</u>

The fine details of epithelial cells; including their nuclei and cytoplasm; clear goblet cells; and their mucus content; cellular junctions; and microvilli on the epithelial surface;



Transverse section of small intestine





6. Rat cheek cells (Epithelial tissue):

Observation of squamous epithelial cells

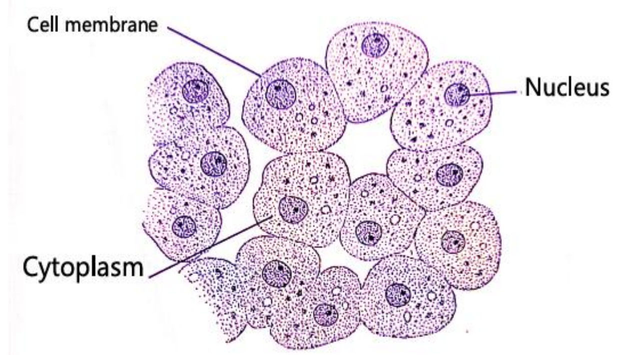
Aim: To examine the structure of squamous epithelial cells from the cheek lining of a rat under different magnifications.

Procedure

1) Gently scrap the inside of a rat's cheek using a clean toothpick to collect epithelial cells.
2) Transfer the collected sample onto a clean glass slide.
3) Add a drop of **methylene blue** or **crystal violet** to stain the cells for better visibility.
4) Place a coverslip over the sample to spread the stain evenly and remove air bubbles.
5) Examine the slide under different magnifications to study cell structure in detail.

Observations at different magnifications

<u>Low Power (4x or 10x Objective Lens)</u>
The overall arrangement of squamous epithelial cells as irregular, thin, and flat structures forming a continuous layer.
<u>Medium Power (20x or 40x Objective Lens)</u>
Individual cells become clearer; centrally located nucleus in each cell is visible; thin cell membrane enclosing the cytoplasm is distinguishable.
<u>High Power (100x Oil Immersion Objective Lens)</u>
Detailed observation of the nucleus, appearing darker due to the stain; cytoplasm and thin, flattened shape of the cells are more defined; cell boundaries and any overlapping structures are clearly visible.



Cheek cells: *Typical cheek cells are flat, irregularly shaped epithelial cells with a thin cell membrane, cytoplasm, and a prominent round nucleus.*
They are usually loosely arranged and lack a cell wall and chloroplasts, indicating that they are animal cells.

7. Rat skeletal muscle tissue

Aim: To examine the structure of skeletal muscle tissue, focusing on muscle fibres, striations, and nuclei under different magnifications.

Procedure

1. Obtain a thin section of rat skeletal muscle tissue.
2. Stain the tissue section using **eosin**, which enhances visibility of muscle fibres and structural details.

3. Place the stained tissue on a clean glass slide, cover it with a coverslip, and ensure there are no air bubbles.
4. Examine the slide under different magnifications to study the muscle fibres in detail.

Expected Observations at Different Magnifications

Low Power (4x or 10x Objective Lens)

Muscle fibres appear as long, cylindrical structures arranged parallel to each other.

Medium Power (20x or 40x Objective Lens)

Striations (alternating light and dark bands) in the muscle fibres become visible; multiple nuclei can be seen positioned at the periphery of the muscle fibres.

High Power (100x Oil Immersion Objective Lens)

Sarcomeres (functional contractile units) are more clearly defined within the striated pattern; detailed structures of the muscle fibres, including the arrangement of actin and myosin filaments; can be observed; multiple peripheral nuclei are distinctly visible.

8. Rat cardiac muscle tissue:

Aim: To examine the structure of rat cardiac muscle tissue, focusing on striations, intercalated discs, and branched fibres under different magnifications.

Procedure

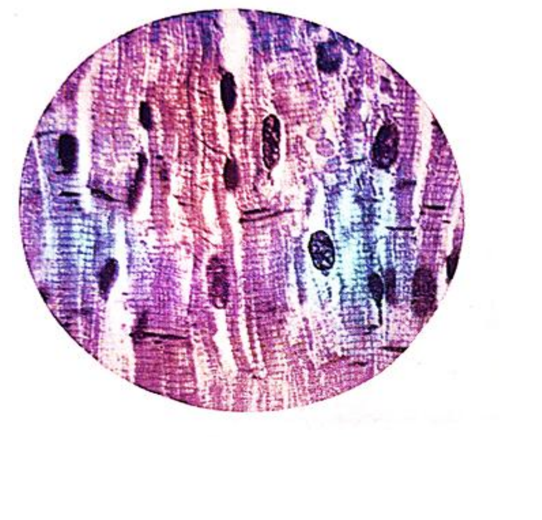
1) Obtain a thin transverse section of rat heart tissue.
2) Apply hematoxylin and eosin (H&E) stain to enhance the visibility of muscle fibres, striations, and intercalated discs.
3) Place the stained tissue on a clean glass slide, cover it with a coverslip, and ensure even spreading to avoid air bubbles.
4) Examine the prepared slide under different magnifications to analyse the structural details of cardiac muscle.

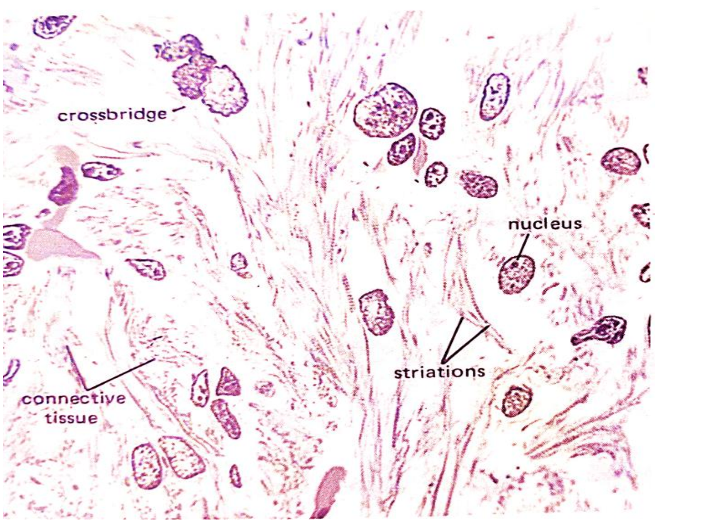
Expected observations at different magnifications

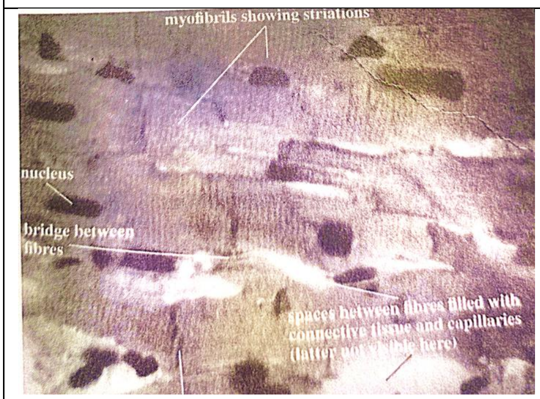
Low Power (4x or 10x Objective Lens)

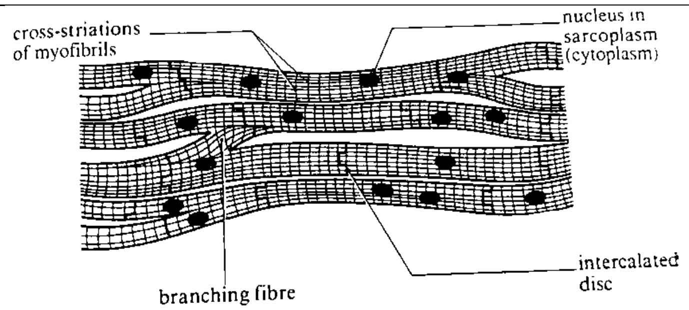
The overall arrangement of cardiac muscle fibres is visible; Muscle fibres appear branched and interconnected; forming a network-like structure;

<u>Medium Power (20x or 40x Objective Lens)</u>
Striations; alternating light and dark bands within the cardiac muscle fibres become visible; Intercalated discs (dark, dense lines) appear at regular intervals; connecting adjacent muscle fibres.
<u>High power (100x Oil Immersion Objective Lens)</u>
Striations are more defined, showing the arrangement of contractile proteins; Intercalated discs are more distinct, revealing their role in cell-to-cell communication and synchronized contraction; central nuclei within the muscle fibres become clearly visible;

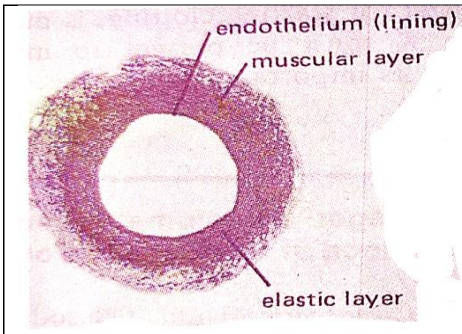


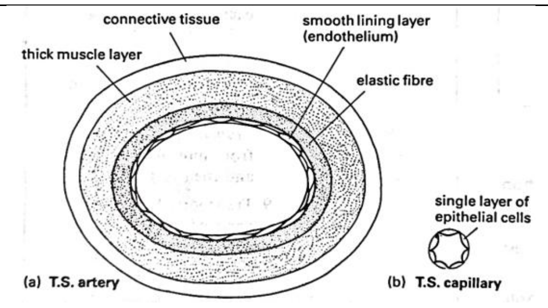


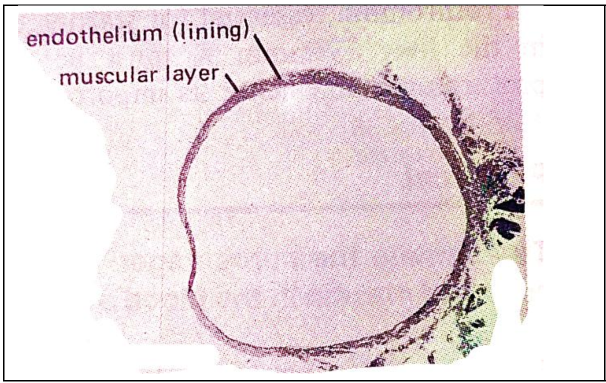


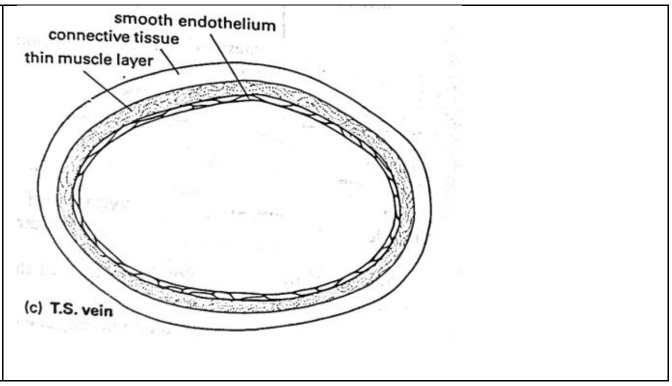


Blood vessels









9. Rat adipose tissue:

Aim: To examine the microscopic structure of rat adipose tissue, focusing on adipocytes (fat cells), nuclei, and lipid storage vacuoles under different magnifications.

Procedure

Obtain a thin section of rat adipose tissue.

Stain the tissue section using hematoxylin and eosin (H&E) to enhance the visibility of adipocytes and their structural components.

Place the stained tissue on a clean glass slide, cover it with a coverslip, and ensure even spreading to avoid air bubbles.

Examine the slide under different magnifications to study the structural details of adipose tissue.

Expected observations at different magnifications

<u>Low Power (4x or 10x Objective Lens)</u>

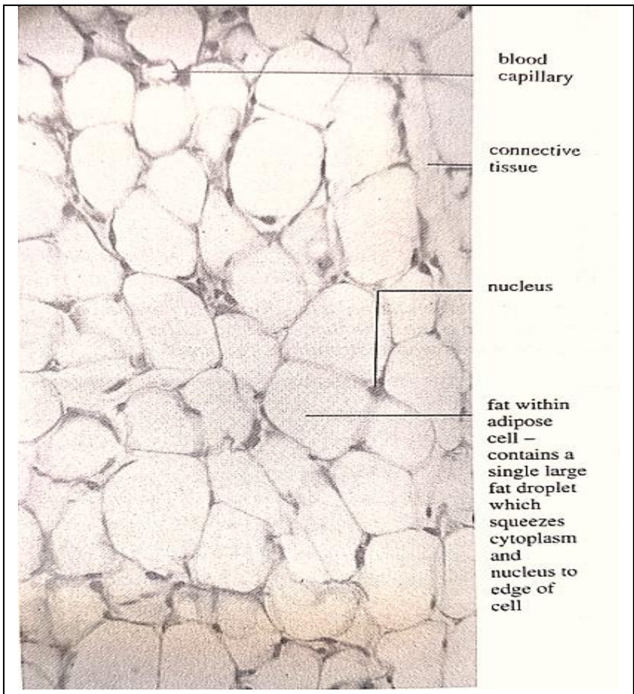
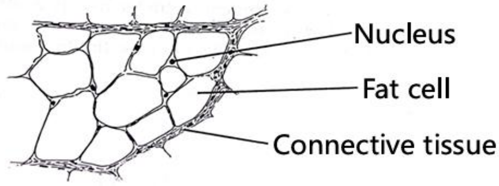
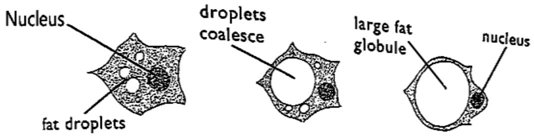
The tissue is a loose arrangement of large, round adipocytes; The adipocytes are closely packed; forming a honeycomb-like pattern; each cell contains a large, unstained lipid droplet; the nucleus to the periphery.

<u>Medium Power (20x or 40x Objective Lens)</u>

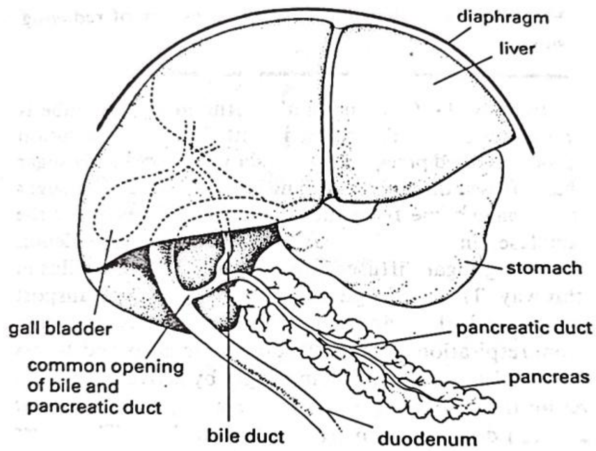
Individual adipocytes are clearly distinguishable; thin cytoplasmic membrane surrounding each adipocyte is more visible; nuclei appear flattened and pushed to the edges of the cells due to the large lipid droplet.

<u>High Power (100x Oil Immersion Objective Lens)</u>

The fine details of adipocytes are observed, including the thin cytoplasmic layer surrounding the lipid droplet; The peripheral nucleus is clearly seen, sometimes appearing as a small, dark-stained structure along the edge of the cell; The absence of visible organelles in most of the cytoplasm due to the dominance of lipid storage is evident.

	*Small portion of the adipose tissue*  *Accumulation of fat in fat cell* 

Rat Liver issue (Liver cells):
Aim: To examine the microscopic structure of rat liver tissue, focusing on hepatocytes (liver cells), sinusoids (blood vessels), and lobular organization under different magnifications.

Procedure 1) Obtain a thin section of rat liver tissue. 2) Stain the tissue section using hematoxylin and eosin (H&E) to enhance the visibility of hepatocytes, sinusoids, and other cellular structures. 3) Place the stained tissue on a clean glass slide, cover it with a coverslip, and ensure even spreading to avoid air bubbles. Examine the slide under different magnifications to study the detailed structures of liver tissue.	 *The liver is a large, vital organ that detoxifies blood, produces bile, stores energy, and helps in metabolism. It is located in the upper right abdomen, just below the diaphragm and mostly under the right ribs.*

Expected observations at different magnifications

<u>Low Power (4x or 10x Objective Lens)</u>

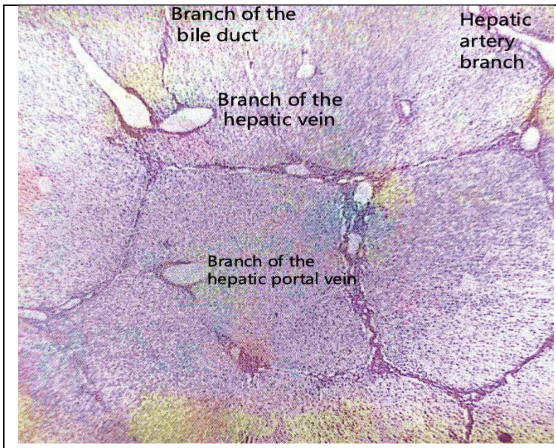
The liver appears as organized lobules; which are hexagonal in shape; functional units of the liver; Each lobule contains central veins; surrounded by hepatocytes and sinusoids;

<u>Medium Power (20x or 40x Objective Lens)</u>

Hepatocytes appear as polygonal cells with a centrally located nucleus; Sinusoids (small blood vessels) are visible as spaces between the rows of hepatocytes; Kupffer cells (specialized macrophages) may be seen lining the sinusoids;

<u>High Power (100x Oil Immersion Objective Lens)</u>

Hepatocytes are clearly visible with their prominent nucleus; and cytoplasm; Sinusoids appear as small; irregularly shaped spaces; endothelial lining of sinusoids and scattered Kupffer cells are more distinct.



The hepatic lobule of a rat liver is organized around a central hepatic vein, with a portal triad comprising the hepatic portal vein, hepatic artery, and bile duct at its outer corners. It consists of radiating rows of hepatocytes separated by blood-filtering sinusoids and bile channels, allowing the liver to simultaneously process nutrients and produce bile.

Adaptations of liver tissue to Its functions

Observed Structure	**Adaptation to Function**
Hepatocytes (liver cells)	Large cells with many mitochondria for energy production, and smooth/rough ER for detoxification and protein synthesis.
Central nucleus in hepatocytes	Controls cell activities and allows for rapid regeneration of liver tissue.
Sinusoids (blood vessels)	Wide and permeable to allow efficient exchange of nutrients, oxygen, and toxins between blood and liver cells.
Lobule arrangement	Hexagonal lobules for efficient filtration and metabolic processing of blood.

Kupffer cells in sinusoids	Specialized macrophages that remove bacteria, toxins, and old red blood cells from the blood.

Rat nerve tissue (Neurons)
Aim: To examine the microscopic structure of rat nerve tissue, focusing on neurons (axon, dendrites) and neuroglial cells under different magnifications.
Procedure
1) Obtain a thin section of rat spinal cord or brain tissue.
2) Stain the tissue section with toluidine blue to enhance the visibility of neurons and neuroglial cells.
3) Place the stained tissue on a clean glass slide, cover it with a coverslip, and ensure even spreading to avoid air bubbles.
4) Examine the slide under different magnifications to study the detailed structures of nerve tissue.

Expected observations at different magnifications
Low Power (4x or 10x Objective Lens)
The grey and white matter in the spinal cord or brain tissue is visible; large cell bodies are scattered within the grey matter while the white matter appears as lighter regions.
Medium Power (20x or 40x Objective Lens)
Neurons appear as large cells with a clearly visible nucleus; dendrites and axons extending from the neuron cell body start to become distinguishable; neuroglial cells, which are smaller than neurons, are visible surrounding the neurons, providing support.
High Power (100x Oil Immersion Objective Lens)
Detailed neuron structure is visible, including dendrites (short, branching projections) and the axon (long extension for impulse conduction); Nucleolus within the neuron's nucleus is clearly seen, indicating active protein synthesis; neuroglial cells, which appear smaller and darker, are observed supporting and nourishing the neurons.

Adaptations of Nerve Tissue to Its Functions

Observed Structure	Adaptation to Function
Neuron (large cell body with nucleus)	Houses the nucleus and organelles for high metabolic activity.
Dendrites (branched extensions)	Increase surface area to receive signals from other neurons.
Axon (long projection)	Conducts nerve impulses over long distances.
Myelin sheath (if present in white matter)	Insulates the axon, speeding up impulse transmission.
Neuroglial cells (small supportive cells)	Provide nutrition, protection, and structural support to neurons.
Gray matter (contains neuron cell bodies)	Processes and integrates nerve signals.
White matter (contains myelinated axons)	Transmits signals between different parts of the nervous system.

Rat kidney tissue (Renal tubules):

Aim: To examine the microscopic structure of rat kidney tissue, focusing on renal tubules, glomeruli, and surrounding epithelial cells under different magnifications.

Procedure

1) Obtain a thin section of rat kidney tissue.
2) Stain the tissue section using Periodic Acid-Schiff (PAS) stain, which highlights the basement membranes of renal tubules and glomeruli.
3) Place the stained tissue on a clean glass slide, cover it with a coverslip, and ensure even spreading to avoid air bubbles.
4) Examine the slide under different magnifications to study the detailed structures of kidney tissue.

Expected observations at different magnifications

<u>Low Power (4x or 10x Objective Lens)</u>

The general organization of kidney tissue is visible, including distinct areas of the cortex and medulla; the glomeruli appear as dense, round structures in the renal cortex; renal tubules appear as small circular or elongated structures scattered throughout the section.

Medium Power (20x or 40x Objective Lens)

Glomeruli are more clearly defined as a dense cluster of capillaries, surrounded by Bowman's capsule; Renal tubules show a central lumen surrounded by cuboidal epithelial cells.

Differences between proximal convoluted tubules (PCT) and distal convoluted tubules (DCT) become clearer:

PCT has a brush border (microvilli), making its lumen appear irregular while the DCT has a clearer, more defined lumen.

High Power (100x Oil Immersion Objective Lens)

Fine details of the glomerulus are visible, showing capillary loops surrounded by podocytes; renal tubule epithelial cells show distinct nuclei and cell boundaries; brush border in PCT is visible, indicating its role in absorption; the basement membrane of the tubules is enhanced by PAS staining, emphasizing its structural role.

Adaptations of kidney tissue to its functions

Observed Structure	**Adaptation to Function**
Glomerulus (capillary network)	*Provides high surface area for filtration of blood.*
Bowman's capsule	*Collects filtrate from the glomerulus for further processing.*
Proximal convoluted tubule (PCT)	*Lined with cuboidal cells with microvilli to increase surface area for reabsorption of nutrients, ions, and water.*
Distal convoluted tubule (DCT)	*Lined with cuboidal cells without microvilli, specialized for selective secretion and reabsorption.*
Renal tubule lumen	*Allows passage of filtrate and urine toward the collecting ducts.*
Cuboidal epithelium in tubules	*Provides structural support and active transport functions for reabsorption and secretion.*

Bacterial Cell

Aim: To examine the structure and shape of bacterial cells using Gram staining or Methylene blue staining, focusing on Staphylococcus aureus (cocci) and Escherichia coli (bacilli/rod-shaped bacteria) under different magnifications.

Procedure

1) Obtain a bacterial culture of Staphylococcus aureus (spherical cocci) or Escherichia coli (rod-shaped bacilli).
2) Transfer a small amount of the bacterial sample onto a clean glass slide.
3) Spread it into a thin smear using a sterile loop.
4) Pass the slide briefly through a flame to fix the bacteria onto the slide, preventing them from washing off during staining.
5) Apply either: Gram Stain (Differentiates Gram-positive & Gram-negative bacteria): Apply Crystal Violet, Iodine, Alcohol, and Safranin in sequence.
6) Staphylococcus aureus (Gram-positive) stains purple.
7) Escherichia coli (Gram-negative) stains pink/red.
8) Methylene Blue (Simple stain to enhance bacterial visibility).
9) Examine the stained smear under different magnifications to study bacterial morphology.

Expected Observations at Different Magnifications

<u>Low power (10x objective lens)</u>

*Bacteria appear as **tiny scattered dots** but lack clear shape and arrangement; the stain colour is visible, but **individual bacteria** are not well-defined.*

<u>Medium power (40x objective lens)</u>

Bacterial shapes become more distinct: Staphylococcus aureus appears as small spherical clusters (grape-like formations). Escherichia coli appears as short rod-shaped cells arranged randomly or in pairs; Some contrast in staining can be observed, especially in Gram-stained samples.

<u>High Power (100x Oil Immersion Objective Lens)</u>

Detailed bacterial morphology is clearly visible:

Staphylococcus aureus (Gram-positive) appears as round purple clusters. Escherichia coli (Gram-negative) appears as pink/red rod-shaped cells. Cell arrangements (clusters, chains, or single cells) are *distinctly seen. Gram stain reaction helps in distinguishing Gram-positive (thick cell wall) vs. Gram-negative (thin cell wall) bacteria.*

Adaptations of bacterial structure to function

Observed Structure	**Adaptation to Function**
Coccus shape (Staphylococcus aureus)	*Compact spherical shape allows resistance to desiccation and immune defence mechanisms.*
Rod shape (Escherichia coli)	*Elongated form increases surface area for nutrient absorption and faster division.*
Gram-positive (thick peptidoglycan wall in Staphylococcus aureus)	*Provides structural strength and resistance to environmental stress.*
Gram-negative (outer membrane in Escherichia coli)	*Acts as a protective barrier, making it more resistant to antibiotics.*
Cluster formation (Staphylococcus aureus)	*Enhances survival in host tissues, making it more resistant to immune attacks.*
Random or paired arrangement (Escherichia coli)	*Allows rapid movement and adaptation to various environments.*

PLANT TISSUES